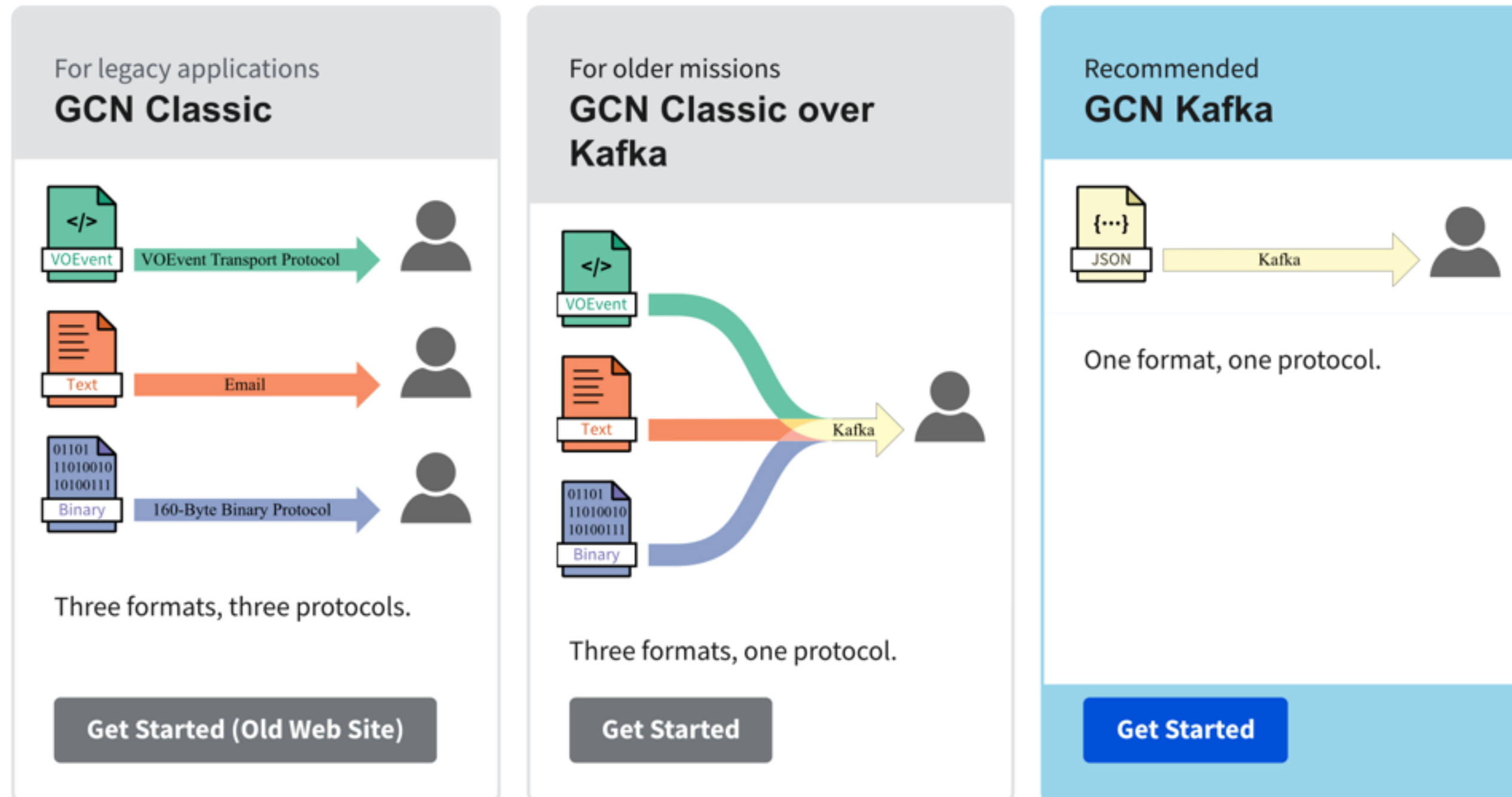


What's new about GCN



A **protocol** defines **how data and alerts are communicated** between systems — that is, the *language, format, and transport method* used to send alerts from detectors to subscribers

WHY APACHE KAFKA

- **Publish/subscribe model:** producers (missions) publish events, consumers (observers, brokers) subscribe to topics
- **Real-time streaming** with millisecond-scale latency
- **Reliable delivery and persistence** — clients can reconnect and replay missed messages
- **Scalable** — supports hundreds of missions and thousands of subscribers
- **Uniform JSON schema** — no need for mission-specific parsers

What's new about GCN

	GCN Classic	GCN Classic over Kafka
 Self-service	NO. Users need to contact administrator in order to make account and subscription changes	YES. Manage your own account and subscription settings through the web site
 Open standards	NO. Notices are sent using three custom protocols	YES. Notices are sent using one standard protocol, Apache Kafka
 Open source	NO. Custom software needed to receive notices	YES. Receive notices using open-source software
 Highly available	NO. Notices are broadcast by a single server	YES. Notices are broadcast by a cluster of highly-available Kafka brokers in the cloud
 Secure	NO. Notices are sent as plaintext	YES. Notices are protected with SSL/TLS